Pseudo-Relevance Feedback for Dense Sequential Recommendation

Giulio D'Erasmo[1], Filippo Betello[1], Federico Siciliano[3], Nicola Tonellotto[2] and Fabrizio Silvestri[1]

[1]Sapienza University of Rome, Rome, Italy
[2]University of Pisa, Pisa, Italy
[3]National Institute of Geophysics and Volcanology, Rome Italy

Abstract

Sequential Recommender Systems (SRSs) capture dynamic user preferences effectively, but they rest on the assumption that interaction histories are a clean signal. In practice implicit feedback is noisy: exploratory clicks and one-off actions distort the learned user representation. Drawing on classical Information Retrieval principles, we adapt the Rocchio relevance feedback algorithm to the latent space of sequential recommenders. Our method refines the user representation by shifting the sequence embedding towards a pseudo-positive view, the recent interaction window, and away from a pseudo-negative view built from non-interacted items. Departing from the traditional post-hoc application of relevance feedback, we make the update rule fully differentiable and optimise its coefficients end-to-end, turning Rocchio from a static heuristic into a learned component. We instantiate the mechanism as a shared-encoder variant, which acts as a self-regulariser, and a decoupled-encoder variant designed for disentangled representations; both are model-agnostic and integrate into existing SRS backbones. Experiments on three benchmarks show consistent gains, up to $+6.03\%$ NDCG@5 over the strongest baseline and up to $+9.61\%$ MRR@5 over the corresponding backbone. Our code is available at https://anonymous.4open.science/r/ROCCHIO_recsys-8CD8.

Keywords

Recommender Systems, Relevance Feedback, Rocchio, Neural Methods

1. Introduction

Recommender systems shape user experience across e-commerce, entertainment and social media [1, 2, 3], and Sequential Recommender Systems (SRSs) do so by exploiting the temporal structure of user interactions [4**?**]. SRSs have evolved rapidly, progressing from Markov chains [5] and Recurrent Neural Networks such as GRU4Rec [6] to Transformer-based architectures. State-of-the-art models such as SASRec [7] and BERT4Rec [8] leverage self-attention to capture long-term dependencies in user behaviour. Despite their expressive power, these models operate under a fragile assumption: that interaction histories are a reliable signal [9]. Implicit feedback is inherently noisy, containing accidental clicks, exploratory behaviour and one-off actions that do not reflect the true user preference [10].

To mitigate the impact of unreliable interaction data, the community has largely adopted Contrastive Learning (CL). Pioneering works such as CL4SRec [11] and CoSeRec [12] maximise the agreement between a sequence and its augmented views (cropped, masked or shuffled histories), treating derived sub-sequences as positive signals in order to anchor the user representation to a core intent. A critical distinction, however, lies in the mechanism of improvement. CL methods rely on stochastic augmentations to enforce invariance, teaching the model to ignore noise; our approach is instead explicitly corrective. Rather than merely tolerating perturbations, we leverage the Rocchio mechanism to actively refine the user representation, shifting the embedding towards informative patterns and away from irrelevant noise in the latent space.

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gderasmo@diag.uniroma1.it (G. D'Erasmo); betello@diag.uniroma1.it (F. Betello); siciliano@diag.uniroma1.it (F. Siciliano); nicola.tonellotto@unipi.it (N. Tonellotto); fsilvestri@diag.uniroma1.it (F. Silvestri)
0009-0002-2287-051X (G. D'Erasmo); 0009-0006-0945-9688 (F. Betello); 0000-0003-1339-6983 (F. Siciliano); 0000-0002-7427-1001 (N. Tonellotto); 0000-0001-7669-9055 (F. Silvestri)
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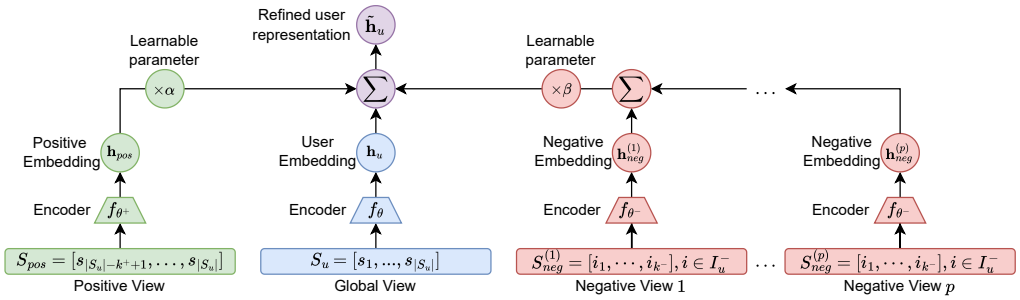


Figure 1: Overview of the proposed framework. The user's full sequence (S_u), the recent positive window (S_{pos}) and sampled negative interactions (S_{neg}) are processed by parallel encoders; the resulting embeddings are aggregated with learnable weights (α, β) into the refined user representation $\tilde{\mathbf{h}}_u$ used for item scoring.

We reach this mechanism by revisiting the problem from the perspective of classical Information Retrieval (IR). If the item catalogue is viewed as a document collection and the user history as a query, recommendation becomes a retrieval process. In IR, ambiguous queries are refined via Relevance Feedback [13, 14]: the query vector is adjusted to move closer to relevant documents and away from irrelevant ones [15], as formalised by the Rocchio algorithm [16]. While this principle remains effective in modern dense retrieval [17, 18, 19, 20], its application to SRSs is unexplored. The closest previous attempt [21] is limited to static collaborative filtering and, crucially, relies on explicit ratings to define the positive and negative sets. That assumption does not transfer to SRSs, where feedback is implicit, sequential and lacks explicit negative labels.

We therefore adapt the Rocchio mechanism to the latent space of SRSs by generating *pseudo-relevance* signals directly from the sequence: the most recent sub-sequence acts as the positive view, while a sample of non-interacted items acts as the negative view. We introduce two architectural variants: a *Shared-Encoder* framework, which reuses a unified backbone across views to maximise parameter efficiency, and a *Decoupled-Encoder* framework, which assigns distinct, jointly-learned backbones to each view in order to disentangle their representations. To summarise, our contributions are:

1. We bridge classical IR and modern deep learning by adapting Rocchio relevance feedback to SRSs, with feedback coefficients that are learned end-to-end rather than fixed a priori.
2. We introduce a mechanism to generate pseudo-positive and pseudo-negative views from interaction histories, enabling the application of Rocchio in implicit feedback settings.
3. We demonstrate that our framework is model-agnostic and consistently improves the robustness of state-of-the-art SRS models against interaction noise on three benchmarks.

2. Methodology

Problem formulation. The objective of an SRS is to model dynamic user preferences from historical behaviour in order to forecast future interactions. Consider a set of n users $\mathcal{U} \subset \mathbb{N}^+$ and a set of m items $\mathcal{I} \subset \mathbb{N}^+$. The interaction history of each user $u \in \mathcal{U}$ is a chronological sequence $S_u = [s_1, \ldots, s_{|S_u|}]$, with $s_i \in \mathcal{I}$. An SRS $\mathcal{M}$ processes a truncated sequence of maximum length L, denoted $S_u^L = [s_1, \ldots, s_L]$, and recommends the next item s_{L+1}: the output $r^{L+1} = \mathcal{M}(S_u^L) \in \mathbb{R}^m$ is a distribution of ranking scores over the item set. Items in S_u are regarded as positives and the remaining ones as negatives [22], $\mathcal{I}_u^- = \mathcal{I} \setminus S_u$, so the problem amounts to ranking s_{L+1} above items drawn from $\mathcal{I}_u^-$. Internally, an encoder f_θ with learnable parameters θ maps the input sequence into a hidden representation

$$\mathbf{h}_u = f_\theta(S_u^L) \in \mathbb{R}^d,$$

which is then used to compute ranking scores for all candidate items via a dot product or a decoding layer.

Rocchio-enhanced representation. Standard encoders assume that S_u^L is a clean signal. Inspired by the Rocchio algorithm [16], we instead refine $\mathbf{h}_u$ by explicitly pulling it towards a *positive centroid* and pushing it away from a *negative centroid*. Since explicit relevance labels are unavailable in SRSs, we generate *pseudo-views* from the sequence itself, defining three distinct views of user intent:

1. **Global view (query):** the original input sequence S_u^L, representing the full context.
2. **Positive view (local intent):** a sub-sequence $S_{pos} = [s_{|S_u|-k^+ +1}, \ldots, s_{|S_u|}]$ containing only the most recent k^+ interactions. This view captures the immediate user drift, assuming that recent items are stronger indicators of current intent [23].
3. **Negative view (noise):** a sequence S_{neg} built by sampling k^- items from the non-interacted set $\mathcal{I}_u^-$, representing a distant preference profile. To reduce variance we sample p different negative sequences $\{S_{neg}^{(j)}\}_{j=1}^{p}$ and average their representations.

Rocchio update rule. We integrate these views into a unified representation through a learnable Rocchio update. Let $\mathbf{h}_{pos} = f_{\theta^+}(S_{pos})$ and $\mathbf{h}_{neg}^{(j)} = f_{\theta^-}(S_{neg}^{(j)})$. The refined user vector $\tilde{\mathbf{h}}_u$ is computed as

$$\tilde{\mathbf{h}}_u = \mathbf{h}_u + \alpha \cdot \mathbf{h}_{pos} + \beta \cdot \frac{1}{p} \sum_{j=1}^{p} \mathbf{h}_{neg}^{(j)}. \tag{1}$$

Unlike the classical algorithm, which uses fixed hyperparameters, we define $\alpha, \beta \in \mathbb{R}$ as *learnable scalar parameters*. This allows the model to adaptively determine how much to trust the recent intent and how aggressively to filter noise during training.

Architectural variants. We investigate two ways of implementing the encoders. In `Rocchio-SE` (shared encoder) all views share the same parameters, $\theta = \theta^+ = \theta^-$. This acts as a self-regularisation mechanism, in which the model learns to encode coherent sequences and noise patterns within the same latent space. In `Rocchio-DE` (decoupled encoders) each view has its own parameters, $\theta \neq \theta^+ \neq \theta^-$, so that the main encoder focuses on the global context while θ^+ specialises in short-term interests and θ^- explicitly characterises random noise distributions. The entire model is trained end-to-end by minimising the cross-entropy loss over the next-item prediction task.

3. Experimental Settings

Datasets. We evaluate on three benchmark datasets widely employed for next-item recommendation, covering different domains: MovieLens-1M (ML-1M, movies), Amazon Beauty (Beauty, e-commerce) and Foursquare New York City (FS-NYC, point-of-interest). Following standard preprocessing protocols [7, 8], we treat all interactions as implicit feedback, discarding review ratings, and remove users and items with fewer than five interactions. We adopt the leave-one-out splitting strategy: the last interaction of each user sequence is used for testing, the second-to-last for validation and the rest for training.

Backbones and baselines. Since our framework is a model-agnostic module, we evaluate it by integrating it into two representative sequential backbones: GRU4Rec [6], a pioneering RNN-based model using Gated Recurrent Units, and SASRec [7], a self-attentive model that uses Transformer layers to capture long-term dependencies. Our evaluation therefore compares the original backbones against their Rocchio-enhanced counterparts (`Rocchio-SE` and `Rocchio-DE`). We additionally report BERT4Rec [8] and two widely adopted contrastive learning approaches, CL4SRec [24] and DuoRec [25]. We consider contrastive learning as orthogonal to our framework, since our module operates independently of specific contrastive losses and augmentation pipelines and is therefore potentially complementary rather than competing.

Table 1
Performance comparison on three benchmark datasets. The best results are in **bold** and the second-best are underlined. "Δ Best Baseline" reports the relative improvement of the best Rocchio-SE variant over the strongest baseline per column; "Δ GRU4Rec" and "Δ SASRec" report the relative improvement of each Rocchio-SE variant over its corresponding backbone.

Model	Amazon Beauty				Foursquare NYC				ML-1M			
	NDCG		MRR		NDCG		MRR		NDCG		MRR	
	@5	@10	@5	@10	@5	@10	@5	@10	@5	@10	@5	@10
GRU4Rec	0.6282	0.6297	0.6444	0.6491	0.4379	0.4524	0.4903	0.5355	0.0989	0.1281	0.1530	0.2439
BERT4Rec	0.6498	0.6503	0.6641	0.6656	0.3295	0.3400	0.3647	0.3970	0.0862	0.1100	0.1329	0.2066
SASRec	0.6478	0.6490	0.6609	0.6648	0.3983	0.4108	0.4571	0.4958	0.1198	0.1510	0.1815	0.2781
CL4SRec	0.6487	0.6497	0.6648	0.6680	0.3923	0.4072	0.4451	0.4922	0.1224	0.1526	0.1849	**0.2788**
DuoRec	0.6500	0.6505	0.6625	0.6641	0.3956	0.4088	0.4497	0.4903	0.1225	0.1517	**0.1869**	0.2785
Rocchio-DE (SASRec)	–	–	–	–	–	–	–	–	–	–	–	–
Rocchio-DE (GRU4Rec)	–	–	–	–	–	–	–	–	–	–	–	–
Rocchio-SE (SASRec)	0.6479	0.6505	**0.6688**	0.6766	0.4318	0.4407	0.4700	0.4977	**0.1232**	**0.1536**	0.1834	0.2776
Rocchio-SE (GRU4Rec)	**0.6529**	**0.6565**	0.6664	**0.6774**	**0.4643**	**0.4786**	**0.5162**	**0.5605**	0.1077	0.1343	0.1677	0.2503
Δ Best Baseline	*+0.45%*	*+0.92%*	*+0.60%*	*+1.41%*	*+6.03%*	*+5.79%*	*+5.28%*	*+4.67%*	*+0.57%*	*+0.66%*	*-1.87%*	*-0.43%*
Δ GRU4Rec	*+3.93%*	*+4.26%*	*+3.41%*	*+4.36%*	*+6.03%*	*+5.79%*	*+5.28%*	*+4.67%*	*+8.90%*	*+4.84%*	*+9.61%*	*+2.62%*
Δ SASRec	*+0.02%*	*+0.23%*	*+1.20%*	*+1.78%*	*+8.41%*	*+7.28%*	*+2.82%*	*+0.38%*	*+2.84%*	*+1.72%*	*+1.05%*	*-0.18%*

Implementation details. The maximum sequence length L is set to 50 for Amazon Beauty and to 200 for both ML-1M and FS-NYC. We optimise the backbone parameters with Adam, a learning rate of $1e^{-3}$ and a batch size of 1024. Backbone hyperparameters (embedding size, number of layers and attention heads) are tuned via grid search on the validation set, adopting the configuration bounds defined in the original literature. For our module, α and β are learned automatically, while the view-generation hyperparameters are tuned within $k^+ \in \{1, \ldots, 10\}$, $k^- \in \{5, \ldots, 50\}$ and $p \in \{1, 3, 5\}$. Hyperparameter optimisation is managed with RayTune via the Asynchronous Successive Halving Algorithm (ASHA) [26, 27], which evaluates configurations at exponentially increasing resource rungs and promotes a configuration only if its intermediate validation metric falls within the top performing fraction of previously observed trials.

Evaluation metrics. We report NDCG@$\{5, 10\}$ (Normalised Discounted Cumulative Gain) and MRR@$\{5, 10\}$ (Mean Reciprocal Rank). In line with recent literature, and to avoid the bias of sampled metrics [22], we perform *full ranking*: for each test user the ground-truth item is ranked against *all* other non-interacted items.

4. Results

Table 1 reports the performance of our Rocchio-enhanced models against five baselines spanning RNN-based (GRU4Rec), self-attentive (SASRec, BERT4Rec) and contrastive-learning (CL4SRec, DuoRec) paradigms.

State-of-the-art performance. Rocchio-SE establishes a new state of the art on *Amazon Beauty* and *Foursquare NYC*, winning all eight metric–cutoff combinations across the two datasets, seven of them with the GRU4Rec backbone. The improvements over the strongest baseline are particularly pronounced on Foursquare NYC, where Rocchio-SE (GRU4Rec) surpasses the runner-up by $+6.03\%$ NDCG@5 and $+5.28\%$ MRR@5. This confirms that explicitly modelling a positive (recent intent) and a negative (noise) view provides a more discriminative signal than augmentation-based contrastive objectives such as CL4SRec and DuoRec. On *ML-1M* the contrastive baselines remain competitive and retain the top positions on MRR, while Rocchio-SE (SASRec) attains the best NDCG at both cutoffs:

gains in ranking quality persist even in the densest benchmark, though the contrastive paradigm keeps a small edge on the reciprocal-rank metric.

Consistency across backbones. A key property of our framework is its model-agnostic nature. The Δ GRU4Rec and Δ SASRec rows show that `Rocchio-SE` yields positive gains in 23 out of 24 metric–dataset combinations, the single regression being a negligible -0.18% on ML-1M MRR@10 for SASRec. The largest relative improvements occur where the backbone is weaker on a given dataset: `Rocchio-SE` (GRU4Rec) gains up to $+9.61\%$ MRR@5 on ML-1M over plain GRU4Rec, and `Rocchio-SE` (SASRec) gains up to $+8.41\%$ NDCG@5 on Foursquare NYC over plain SASRec. The refinement is therefore not tied to a particular architectural family: it complements both recurrent and self-attentive backbones.

Impact of dataset characteristics. The magnitude of the gains correlates with the noise level of the dataset. On *Foursquare NYC*, the sparsest and most exploratory dataset, the improvements over the best baseline are largest (up to $+6.03\%$). *Amazon Beauty*, sparse but with shorter sessions, exhibits smaller yet consistent gains (below $+1.5\%$), reflecting that the baselines are already strong in this regime. On *ML-1M*, the densest and least noisy benchmark, the improvements shrink to fractions of a percent on NDCG and turn slightly negative on MRR. This pattern aligns with our hypothesis: pseudo-relevance feedback acts as a denoising mechanism, and its marginal benefit grows with the noise-to-signal ratio of the interaction history.

Shared vs. decoupled encoders. Comparing the two proposed architectures, `Rocchio-SE` consistently outperforms `Rocchio-DE`. While the decoupled variant provides additional flexibility by allocating separate parameters to each view, the increased capacity appears to encourage overfitting on the smaller benchmarks. Forcing a single backbone to encode the global, positive and negative views in a shared latent space instead acts as an implicit regulariser, and proves to be the more effective design.

Learned coefficients and ablation. The learned parameters of the optimal `Rocchio-SE` configurations empirically validate the geometric intuition of the Rocchio algorithm: in almost all settings the model converges to a positive α and a negative β, pulling the user representation towards the local-intent centroid and pushing it away from the noise centroid. We further observe $|\alpha| > |\beta|$ in most settings, indicating that reinforcing recent intent is the dominant driver of performance while the negative component plays a complementary refinement role. The few exceptions occur on ML-1M, where a larger $|\beta|$ suggests that in dense regimes the backbone already captures the positive signal well and the marginal value of the module comes from noise suppression. The learned window sizes confirm our design assumption: k^+ stays small ($\in [2, 9]$), consistent with the recency hypothesis, while k^- is broader ($\in [16, 34]$) to yield a stable estimate of the negative centroid. An ablation on two degenerate variants of `Rocchio-SE` (GRU4Rec), *Pos-Only* ($\beta = 0$) and *Neg-Only* ($\alpha = 0$), points in the same direction: the full dual-view model is best on Foursquare NYC, confirming that the two views carry complementary signals, whereas on ML-1M the Neg-Only variant is competitive with the full model.

5. Conclusion

We proposed a framework that adapts the Rocchio relevance feedback algorithm to the latent space of sequential recommenders, using pseudo-positive and pseudo-negative views to refine user representations without explicit relevance labels. Experiments on three diverse benchmarks show that the approach consistently improves the robustness of SASRec and GRU4Rec, with the Shared-Encoder variant (`Rocchio-SE`) emerging as the superior configuration: it reaches gains of up to $+6.03\%$ NDCG@5 over the strongest baseline and up to $+9.61\%$ MRR@5 over its corresponding backbone. Our findings confirm that integrating classical relevance feedback into SRSs is an effective, model-agnostic strategy for mitigating exploratory or erratic behaviour in implicit feedback settings. Future work will investigate

adaptive mechanisms to adjust the Rocchio coefficients per user instance, and the integration of the framework into other families of SRSs such as graph neural networks.

Declaration of Generative AI

In the preparation of this paper, generative AI tools, specifically ChatGPT and Claude, were used to assist with grammar checking, language refinement, and the improvement of clarity and coherence. These tools served as supplementary aids in the writing process. All AI-generated suggestions were reviewed, evaluated, and revised by the authors to ensure the accuracy, relevance, and appropriateness of the content. The authors take full responsibility for the final content of this paper, including its accuracy, originality, and academic integrity.

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